FlightPlan Hackathon Brief

From the conference version to a bedside version. Background for hackathon participants.
Cincinnati Children's Hospital Medical Center

1. What FlightPlan is

FlightPlan is a software tool built at Cincinnati Children's to help a care team understand the whole course of a long, complicated hospital stay. It turns a patient's admission into one picture over time: where the patient has been, how sick they were at each point, what procedures and treatments happened, what the team decided, and what still needs follow-up.

The name comes from aviation. A pilot's flight plan shows the origin, the route, the waypoints, the current position, the hazards and the destination. FlightPlan does the same for a hospital stay. The approach grew out of work applying the aviation "threat and error" safety model to pediatric heart surgery (Hickey and colleagues, *Annals of Thoracic Surgery*, 2017, as cited in the FlightPlan beta user guide).

The first use is children having heart surgery or heart catheterization (a procedure done through a thin tube passed into the heart through a blood vessel) in the Cincinnati Children's Heart Institute. FlightPlan does not replace the electronic health record (EHR), the hospital's main medical record system, and it does not make clinical recommendations. It is a shared picture the whole team can look at together.

Today FlightPlan is used mainly away from the bedside:

- **FlightPlan conference.** A weekly meeting, also called Performance Rounds, where cardiology and intensive care physicians, surgeons, anesthesia staff, operating room staff, nurse practitioners, fellows and nursing leaders review recent patients together. The stated aim, in the beta user guide, is "blameless but accountable" learning from both successes and problems.
- **Handoffs.** Physicians and nurses use it at shift change to explain how a patient's stay has gone and whether the patient is meeting expected milestones.
- **Two safety programs that came out of it. CrossCheck** flags vulnerable patients before major procedures. **Course Correction** sets patient-specific thresholds for escalating care quickly when a patient drifts from the expected recovery path.

In this brief, the **conference version** means FlightPlan as it is used today: a web page shown on a large screen in a meeting room. The **bedside version** means the new, smaller view this hackathon will design for daily rounds in the patient's room, with the team standing at the bedside and the family often present.

2. What a FlightPlan shows

A FlightPlan page has five parts:

- **Patient information.** Identifiers, current weight, caregiver contact and the CrossCheck flag.
- **Diagnoses and interventions.** The main diagnoses and each surgery or catheterization, with date, time, weight and the care team for that procedure.
- **Care team.** Names grouped by role: surgeon, anesthesia, cardiac intensive care, step-down unit, cardiologist and intensive care attending (the senior physician in charge).
- **Timeline.** The core picture, described below.

- **Notes, images and documents.** Free-text notes sorted by type, plus imaging and review documents attached to the stay.

The timeline reads left to right through the days of the stay. Its rows are levels of risk, from highest at the top to lowest at the bottom:

Row	What it means
Procedure	The patient is in the operating room or the catheterization lab, or on ECMO (a machine that takes over the work of the heart and lungs).
Intubated	On a breathing machine through a tube in the windpipe, at intensive care level.
Extubated	Breathing tube removed, still in intensive care.
ACCU	The step-down unit (acute care cardiology unit), one level below intensive care.
Discharged	Home.

A single colored line moves between these rows. Its height shows how sick the patient is. Its color shows where the patient is: purple for the operating room, pink for the catheterization lab, orange for the cardiac intensive care unit (CICU) and blue for the step-down unit. Shading marks time on ECMO. The beta user guide describes the ideal stay as a stepwise drop in risk from admission to discharge, so a line that climbs back up shows a setback at a glance.

Other layers sit on top of the line:

- **Pathway.** A dashed line for the expected course, so the team can compare what was expected with what happened.
- **Course Correction.** A dotted line tied to the Course Correction program described in section 1.
- **Milestone symbols.** For example, when heart-support medicines (inotropes) were stopped and when chest drainage tubes were removed.
- **Callout boxes.** Short text attached to a point in time, often echocardiogram (heart ultrasound) or catheterization findings, written in clinical shorthand.
- **Ratings.** Small markers under the timeline, one for each phase of care. Each holds a team performance score from 1 to 4 (4 is outstanding, 3 is good, 2 is sub-optimal, 1 is poor) and a colored dot for the patient outcome in that phase (green if the patient followed the ideal course, yellow for a deviation with limited impact, red for a serious one such as cardiac arrest or an unplanned return to surgery).
- **Conference markers.** Vertical lines that appear to mark the days a case conference discussed the patient.

Under the dates, a second row counts days after surgery. "POD" means post-operative day. Long uneventful stretches are compressed and marked with a zigzag. Patients whose stay went exactly as expected, with no unplanned events and all-green ratings, are called "Direct Flights" and are grouped together at the weekly conference so that time goes to the cases with the most to learn from.

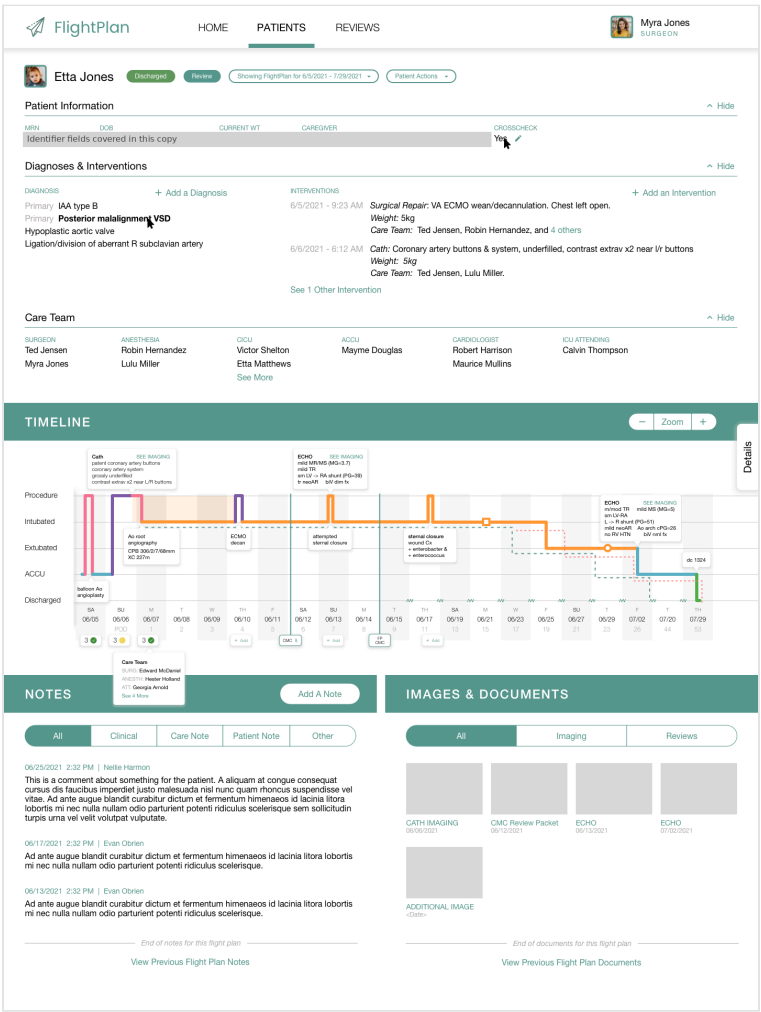
According to the FlightPlan Clinical and Technical Overview, a FlightPlan is prepared before the weekly conference. Key events are compiled from the EHR and checked with clinicians who know the patient. The ratings are entered by each team for its own phase of care.

3. The conference version in two screenshots

Both screenshots come from the FlightPlan design asset set prepared by the human factors firm POMIET (February 2025). The patient, family and staff names are made up and the notes are placeholder text. On the first screenshot, the row of patient identifiers has been covered in this copy.

These show the conference version that will be demonstrated at the hackathon. It works for its purpose. In a conference room, with a large screen and time to talk, a team can walk through a whole stay from left to right. The difficulty is that the bedside is a different setting. The team has a minute or two per patient, possibly on a tablet or other small screen, with the family listening.

Screenshot 1. The full patient page



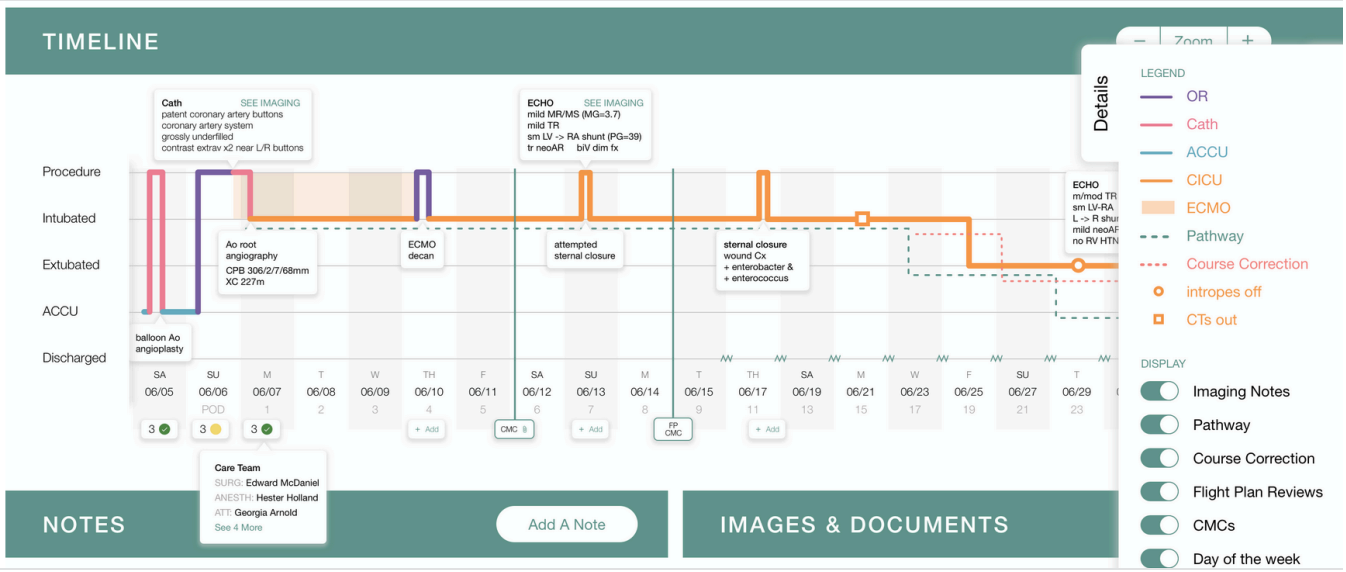
Screenshot 1. The full FlightPlan patient page for a made-up example patient. Design mockup from the POMIET asset set, February 2025.

Why it is hard to use at the bedside:

- **Identifiers lead the page.** The first thing on screen is the medical record number, date of birth and the caregiver's name and phone number. In a room with a family or a roommate present, that is the wrong first thing to show.
- **There is no "now".** The page is organized around the whole stay. Nothing pulls out the patient's current status, what happened overnight, or today's priorities. A clinician has to read the timeline to its right-hand end and work that out.

- **It is long.** Patient information, diagnoses, care team, timeline, notes and images are stacked on one scrolling page sized for a desktop monitor.
- **Capturing decisions is generic.** The ways to add information are "Add a Diagnosis", "Add an Intervention", "Add A Note" and small "+ Add" buttons on the timeline. There is no place to record a decision, assign a task to a person, or set a follow-up.
- **Notes are a plain list.** Notes appear newest first under type tabs. Nothing links a note to the event on the timeline it refers to.
- **One view for everyone.** An attending, a bedside nurse, a consultant and a parent all see the same page.

Screenshot 2. The timeline close up



Screenshot 2. The timeline for the same made-up patient, with the Details panel open. From the FlightPlan Clinical and Technical Overview.

Why it is hard to use at the bedside:

- **The most recent days are hidden.** With the Details panel open, the legend covers the right-hand end of the timeline, which is exactly where today is.
- **A lot to decode.** The legend lists nine symbols and line styles, and the display panel has at least six on and off switches. A newcomer, a consultant or a parent cannot read the picture without a lesson.
- **Color carries the meaning.** Location is shown only by line color, which is hard on small screens, in bright rooms and for people with color vision differences.
- **One line, two meanings.** The same line shows both where the patient is and how sick the patient is. That is efficient once learned and confusing before.
- **Callouts collide.** Boxes of text sit on the line and on each other. A FlightPlan design note on timeline annotations describes this problem: when many events are plotted, fixed text boxes overlap and hide each other and the data.
- **Clinical shorthand.** Callouts read like "sm LV -> RA shunt (PG=39)" and "CPB 306/2/7/68mm". That suits specialists in conference but means little to a family and may be the wrong level of detail for other roles.
- **Ratings are for review, not rounds.** The score and colored dot under each phase are a quality review judgment made after the fact. Whether they belong at the bedside, especially in front of a

family, is an open question.

4. The hackathon challenge

You will see the conference version demonstrated, then talk with the clinicians provided about what in it is useful at the bedside and what is not. Your job is to design the bedside version: something new, built by choosing and combining the most useful parts of the conference version for a smaller screen and a shorter conversation.

The data behind FlightPlan, and how they are collected and stored, will not change. The bedside version is a new way of showing the same information, most likely a subset of what the conference version shows, plus a way to capture what happens on rounds. Spend your effort on what to show and how, not on the data underneath.

The goal is not to redesign the whole application. It is to answer one practical question:

> **What should a clinician see, touch, hide, capture, or hand off during bedside rounds?**

Questions to explore with clinicians:

- What do you need to know in the first 30 seconds at the bedside?
- Which parts of the conference version are essential, optional or distracting during rounds?
- What is safe to show when a family is present?
- What should be captured during rounds: decisions, questions, tasks, contingency plans, consult needs, family updates or follow-up?
- What would make you trust this view during live care?

A strong wireframe shows at least:

- A bedside snapshot: current location, status, key diagnoses, active concerns and today's priorities.
- A compressed version of the stay that tells the story without taking over the conversation.
- A "what changed" area: overnight events, new results, procedures, therapies, changes in location or status, and open questions.
- A way to capture actions during rounds: decisions, tasks, follow-ups and who owns each one.
- A family-present mode: what changes on screen when a patient or family is in the room.

By the end of the hackathon, produce interview findings, a bedside information hierarchy (what goes on the first screen, what is secondary, what stays out), the wireframe, a short design rationale tied to what clinicians said, and a recommendation for what to prototype first.

Keep the concept focused on workflow and communication. FlightPlan is not a clinical decision support tool, so the bedside version should not suggest diagnoses or treatments.

Use only made-up or de-identified examples. Submissions must not include medical record numbers, patient names, dates of birth, real patient screenshots or any other information that could identify a patient.

Sources: FlightPlan Hackathon One-Pager (2026). FlightPlan Bedside Rounds Design Challenge. FlightPlan Clinical and Technical Overview. FlightPlan clinical use summary. Flight Plan Software Beta User Guide. FlightPlan Timeline Annotations design note. POMIET FlightPlan design assets (February 2025).